

Spacetime Is Evidence That Equalization Is Not Yet Complete

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July 2026

(A DISTANCE Interpretation of the Rate of Time and the Scale of Space)

We ordinarily assume that time and space must already exist before the universe can contain anything. Space is imagined as the stage upon which all existence is placed. Time is imagined as the flow through which events proceed in sequence. Objects are thought to exist within space, to move through time, and to exchange energy across the spacetime that surrounds them. DISTANCE reverses this order. Time and space do not first exist so that Momentum may move within them. Time and space emerge as observable expressions of the process through which Momentum is generated and resolved. Spacetime is not the fundamental background of the universe. It is the relational manifestation of unresolved Momentum and unresolved Distance.

To understand this possibility, imagine a universe in which every process of Momentum diffusion and resolution has been completed. All Energy Gaps have disappeared. No effective Momentum remains in any direction. No structure provides a greater opportunity for equalization than any other. Every difference in binding structure and energy distribution that once distinguished one Island from another has been completely equalized. In such a condition, none of the motion described by conventional physics would be observable. Nothing would change into anything else. Nothing would move in any direction. There would be no rotation, vibration, combination, or separation. There would not even be a meaningful basis for distinguishing one event from another, or a before from an after.

It is not sufficient to describe this merely as a state in which everything has stopped. To say that something has stopped already assumes that an object capable of movement exists, that it occupies a position in space, and that it remains motionless for some duration of time. In a

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state of complete equalization, however, these assumptions no longer hold. Without change, there is no distinction between before and after. Without before and after, time cannot be defined. Without distinguishable positions, directions, or relational differences, space cannot be defined either. Time in such a universe would not flow extremely slowly. Time would not exist as a meaningful relation. Space would not be infinitely large or compressed to nothing. There would simply be no distinguishable positions or relationships to which a spatial measure could be applied.

At a deeper level, even the distinction among individual Islands would disappear. An Island is a binding structure differentiated from its surroundings, maintaining or transforming itself through Momentum. If every difference has been resolved, no basis remains for identifying one region as an independent Island. One might say that all Islands have come to possess the same Distance from one another. Yet the more precise conclusion is that Distance itself has become undefined. Distance requires at least two distinguishable states. If no difference remains, there are no distinct terms to compare, no direction, and no separation.

Now imagine that within this completely equalized condition, a minute Energy Gap appears. A localized difference emerges that can be distinguished from the surrounding equilibrium. If this difference acquires a degree of organization and directionality, a new Island is formed. As long as the difference remains unresolved, Momentum appears as the directionality associated with its equalization. The crucial point is that this event does not occur at some location within a preexisting space. Before the new Island emerged, there was no spatial differentiation by which one location could be distinguished from another. The Island does not arise inside space. Rather, the distinction between the Island and its surroundings creates the first spatial relation.

The same is true of time. The equalized state before the emergence of Momentum differs from the state after its emergence. A first difference between states now exists, and a process of resolution begins. Only then can an earlier and a later state be distinguished. Time does not arrive from outside to carry the event forward. The transition among states produces the order we interpret as time. With the emergence of Momentum, three phenomena appear simultaneously. A difference arises between equilibrium and disequilibrium, producing change. The distinction between states produces a temporal relation. The distinction between the new Island and its surroundings produces a spatial relation and directional structure.

Momentum therefore does not move within spacetime. The process through which Momentum is resolved constitutes time and space. Time and space may therefore be understood not as two independent fundamental realities, but as two observational expressions of the same

process of Momentum resolution. Time describes how rapidly states become different. Space describes how easily or with what difficulty relations among states can be reorganized. Time is the measure of the rate of transformation associated with Momentum resolution. Space is the measure of the relational steps and structural difficulty required for that resolution.

Now consider two Islands with different magnitudes of Momentum. One possesses relatively large Momentum, while the other possesses relatively small Momentum. Between them remains an appropriate Energy Gap. This gap prevents them from being either completely disconnected or immediately collapsed into a single structure. An Island with greater Momentum generally contains a greater possibility for change. Its directionality toward equalization is stronger, and its capacity to reorganize internal and external relations is greater. If its internal relationships are relatively simple and its Momentum can be converted directly into state transformation, many changes may occur within a given comparative interval.

From the perspective of an external observer, time within that Island may then appear to proceed rapidly. States are transformed at a high density, and fewer relational stages are required to reach a given degree of change. Space appears relatively small because internal states can be connected and reorganized through strong Momentum, reducing their effective Distance. By contrast, an Island with smaller Momentum undergoes weaker equalization. The magnitude of state transformation within the same comparative interval is smaller, and more relational stages may be required to reorganize one state into another. To an external observer, time within that Island may appear slower, while its internal space may appear larger.

This explanation, however, must not be reduced to the simple proposition that larger Momentum always produces faster time and smaller Momentum always produces slower time. The relative scale of time is not determined by the absolute magnitude of Momentum alone. It also depends on how that Momentum is distributed, redirected, balanced, absorbed into existing binding structures, and converted into effective state transformation. Two Islands with the same overall magnitude of Momentum may possess different temporal scales.

In one Island, the internal structure may be simple, allowing most of the Momentum to produce direct transformation. In another, the same Momentum may pass through countless internal Islands and binding relations. It may be divided among internal rotation, mutual cancellation, structural maintenance, local reorganization, and competing directional tendencies. The total magnitude of Momentum may therefore be identical while the externally observable rate of change differs substantially.

In the structurally simpler Island, Momentum produces state transformation relatively di-

rectly. An external observer detects many changes within a short comparative interval and interprets the Island's time as fast. In the more complex Island, Momentum must pass through many internal relations. Momentum formed in one direction is modified by other internal Momentum. Some of it is canceled. Some is consumed in maintaining existing cohesion. Some appears as rotation or vibration. Only a portion becomes a structural change recognizable from outside. Consequently, even a large amount of total Momentum may produce only a small amount of externally observable transformation. In such a case, the Island's time may appear slower than the observer's time.

The opposite is also possible. An Island may possess relatively little total Momentum, yet if its structure is simple and most of that Momentum is converted directly into state transformation, its temporal scale may appear faster than that of an externally observed, more complex Island. Momentum magnitude and internal complexity must therefore be distinguished. The magnitude of Momentum represents the potential intensity of transformation. Internal complexity determines the relational structure through which that Momentum must pass before becoming observable change. The rate of time observed externally depends on how much of the total Momentum becomes effective within a particular observational relationship. Conceptually, this may be expressed as follows:

$$[\text{Observed temporal rate} \sim \frac{\text{Momentum producing effective state transformation}}{\text{complexity of internal relational readjustment}}]$$

This is not yet a fixed physical equation. It is a conceptual representation of the relationship. Internal complexity does not merely mean the number of internal components. It includes the number of competing Momentum pathways, their directional interactions, the binding structures that must be maintained, the thresholds that must be crossed, and the degree to which internal transformation becomes externally observable.

Effective Momentum is also not a fixed quantity. Which portion of an Island's Momentum becomes effective depends on the relationship being observed. Momentum used to maintain internal cohesion may be ineffective from the perspective of external displacement, yet highly effective from the perspective of structural persistence. Momentum involved in internal rotation may contribute little to linear relocation while remaining essential to the continuing existence of the Island. Time is therefore not a single fixed property contained inside an Island. Time is the comparative result of how rapidly effective Momentum produces distinguishable state differences within a particular relational frame.

An observer inside an Island normalizes time according to the Island's own internal transformations. The observer does not directly experience personal time as unusually fast or slow, because all internal processes, including the observer and the instruments of measure-

ment, participate in the same relational structure. The difference between fast and slow time appears only when the rates of transformation in different Islands are compared. If more change occurs in one Island while a certain amount of change occurs in another, the two Islands possess different temporal scales. Yet neither scale is absolutely normal. Each is formed through its own Momentum structure and internal complexity.

Space can be understood in the same way. The scale of space is not an absolute geometric length already established before relations occur. Space expresses the effective Distance associated with how easily or how difficultly one state can enter into a transformative relation with another. Within a structure of large effective Momentum, relations can be reorganized strongly. Different internal states may be connected through fewer relational stages, and pathways of equalization may be densely formed. In this case, internal space appears small. Within a structure of small effective Momentum, relational transformation is weak. More stages may be required for one state to form an effective relation with another. The states appear farther apart, and internal space appears large.

Yet the scale of space cannot be determined by the total magnitude of Momentum alone. If large Momentum is confined within complex internal bindings and produces little transformation in a specific external relation, effective Distance may remain long in that relation. Conversely, even small total Momentum may produce a very short Distance if it is concentrated strongly in a particular direction. Space is therefore not simply the inverse of total Momentum. Space is the measure of how directly effective Momentum can reorganize a relation in a particular direction.

This interpretation leads naturally to the DISTANCE account of gravity. When two Islands of different Momentum magnitudes maintain an Energy Gap, Distance does not appear equally in every direction. Toward the Island in which large Momentum is concentrated, more pathways of equalization exist. There are more relational opportunities through which unresolved differences can be reduced. From the perspective of a smaller surrounding Island, the effective Distance toward the larger Momentum concentration is shorter than the Distance toward a direction in which Momentum is sparse. Equalization therefore develops naturally toward the direction containing denser pathways of Momentum exchange and relational transformation.

Conventional physics describes this directional tendency as gravitational attraction between masses. One object is said to pull another, causing the two objects to approach. DISTANCE interprets the same observable phenomenon through a different ontology. The larger Island does not pull the smaller Island. The effective Distance is shorter in the direction where

greater Momentum exists, and the directionality of equalization therefore develops toward it.

Energy does not first confront a preexisting spatial separation and then overcome it. The spatial scale itself is formed differently according to the distribution of Momentum. A direction of large Momentum appears near because pathways of equalization are dense. A direction of sparse Momentum appears far because opportunities for effective relational transformation are limited. Gravity is therefore not an independent force. It is the directionality of Distance produced by an asymmetric distribution of Momentum.

Within this interpretation, mass, time, space, and gravity are not separate concepts. A strongly bound structure contains and maintains many internal Momentum relationships. Such a structure forms its own temporal and spatial scales internally. At the same time, it creates a preferred direction of equalization for surrounding Islands and shortens effective Distance in that direction. From an external perspective, motion along this shortened Distance is observed as gravity. Differences in time, differences in space, and gravitational directionality all emerge from the same source. That source is unresolved Distance and the Momentum associated with its resolution.

Even here, however, it would be incorrect to conclude that time inside an Island with large Momentum must always proceed faster than time outside it. Large Islands generally possess complex internal binding structures. Strong Momentum creates a greater potential for transformation, but much of that Momentum may be distributed among the maintenance and reorganization of internal relations. In some cases, large Momentum may lead directly to rapid equalization and therefore to a high observable rate of change. In other cases, the same large Momentum may be dispersed, redirected, and mutually balanced through a complex internal structure, producing a lower observable rate of transformation. There is therefore no simple one-to-one correspondence between Momentum magnitude and temporal rate.

Momentum magnitude is one condition contributing to temporal scale. Internal complexity, directional organization, the proportion of Momentum used to maintain cohesion, and the degree to which Momentum becomes effective in an external relation all participate in determining the relative rate of time. Among Islands of similar complexity, the Island with greater effective Momentum will generally display a higher rate of state transformation. Yet among Islands with the same total Momentum, the more complex internal structure may exhibit a lower externally observable temporal rate.

Complexity does not always slow time, however. A complex structure may organize and amplify internal Momentum in a particular direction, thereby producing greater effective

transformation. Complexity is not an independent force that slows time. Complexity is the structural condition that determines how Momentum becomes effective change. This distinction prevents time and space from being treated as fixed measures existing outside matter and structure.

Each Island forms its own temporal and spatial scales according to its Momentum and internal binding organization. Different Islands do not merely occupy different locations within one identical spacetime. They constitute different scales of transformation and different relational Distances. Human observers experience a single cosmic time and a continuous space because the many different transformation rates and Distances of countless Islands are normalized into a common observational framework. That normalized spacetime is not necessarily the ontology of the universe. It is an interface through which different Momentum systems are compared and interpreted.

When time is fixed as a single common scale and differently organized Momentum structures are observed through it, space appears curved. When space is fixed as a single common scale and different rates of transformation are observed through it, time appears curved. From the perspective of DISTANCE, however, neither an independent time nor an independent space is fundamentally bending. Differences among the rates of Momentum resolution and relational Distances of different Islands are being expressed as temporal dilation or spatial curvature because they are translated into a common scale.

Time and space are not stages prepared in advance so that the universe may contain Momentum. Time is the relative rate at which Momentum transforms structure. Space is the effective Distance through which Momentum reorganizes relations. Gravity is the phenomenon in which the direction of larger effective Momentum is formed as the shorter Distance. Spacetime is the observable result of these differences remaining unresolved.

If all Momentum were resolved, both time and space would disappear. If new Momentum emerged, difference and direction would return, and time and space would emerge with them. The magnitude of Momentum and the organization of internal complexity would then produce different temporal rates and spatial scales. Through the interaction among these different scales, the phenomena called mass, motion, and gravity would become observable. Spacetime is therefore not the container of existence. Spacetime is the expression of the fact that existences have not yet reached one another. Spacetime is evidence that equalization is not yet complete.

An interpretation of spacetime based on Chapter 4 of the essay DISTANCE

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